3.5 Transformations of Exponential Functions - Worksheet

SOLUTIONS

MCR3U

Jensen

1) Describe the transformations that map the function $y = 2^x$ onto each of the following functions...

a) $y = 2^x - 2$

- down 2 units $(y-2)$

b) $y = 2^{x+3}$

- left 3 units $(x-3)$

c) $y = 4^x$

$= 2^{2x}$

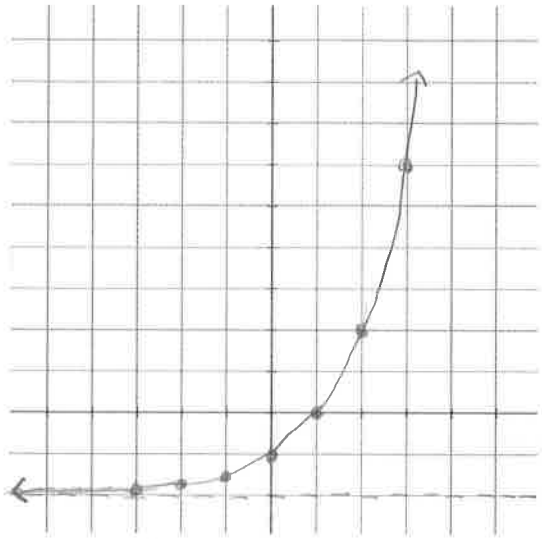
- horizontal compression by a factor of $\frac{1}{2}$ $\left(\frac{x}{2}\right)$

d) $y = 3(2^{x-1}) + 1$

- vertical stretch by a factor of 3 $(3y)$
- right 1 $(x-1)$
- up 1 $(y+1)$

2) Create a sketch of each graph for each equation in question 1. (a table of values may help)

a)



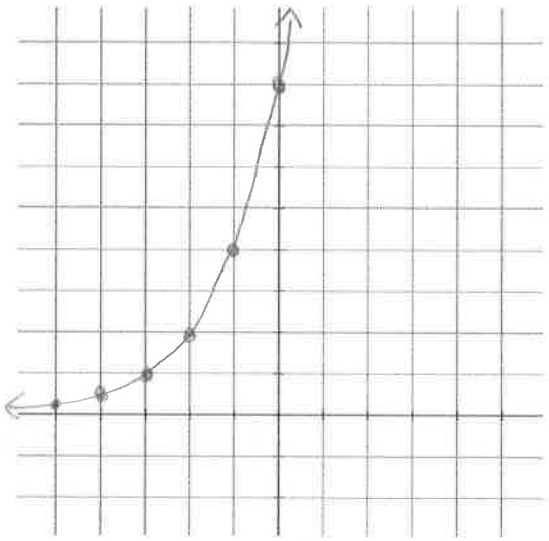
$y = 2^x$

x	y
-3	$\frac{1}{8}$
-2	$\frac{1}{4}$
-1	$\frac{1}{2}$
0	1
1	2
2	4
3	8

$y = 2^x - 2$

x	$y-2$
-3	-1.875
-2	-1.75
-1	-1.5
0	-1
1	0
2	2
3	6

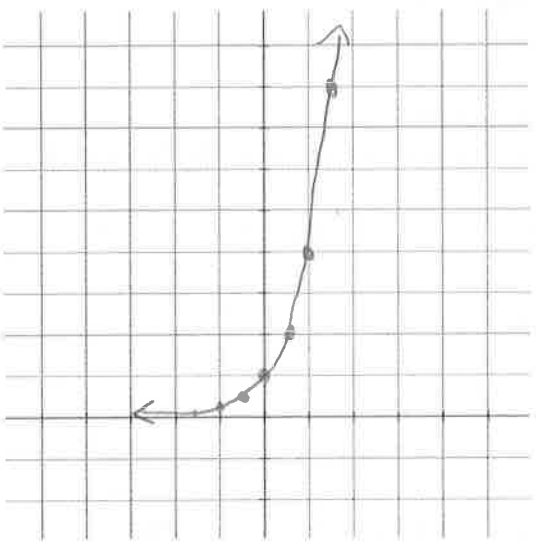
b)



$y = 2^{x+3}$

$x-3$	y
-6	0.125
-5	0.25
-4	0.5
-3	1
-2	2
-1	4
0	8

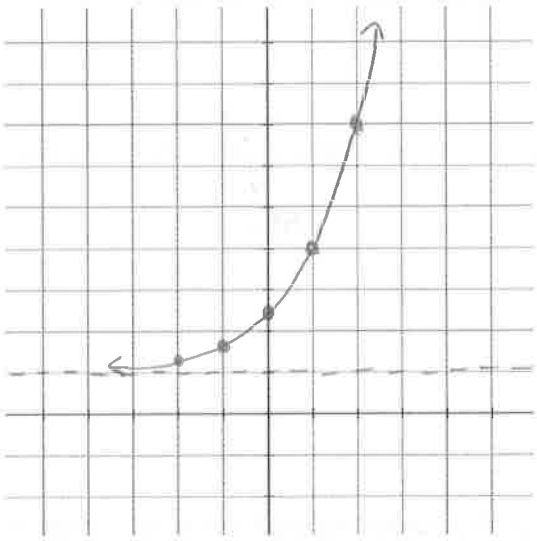
c)



$y = 2^{2x}$

$\frac{x}{2}$	y
-1.5	0.125
-1	0.25
-0.5	0.5
0	1
0.5	2
1	4
1.5	8

d)



$x+1$	$3y+1$
-2	1.375
-1	1.75
0	2.5
1	4
2	7
3	13
4	25

3) Write the equation for the function that results from each transformation applied to the base function $y = 5^x$.

a) translate down 3 units

$y = 5^x - 3$

b) shift right 2 units

$y = 5^{x-2}$

c) translate left $\frac{1}{2}$ unit

$y = 5^{x+0.5}$

d) shift up 1 unit and left 2.5 units

$y = 5^{x+2.5} + 1$

4) Describe the transformations that map the function $y = 8^x$ onto each function.

a) $y = \left(\frac{1}{2}\right)8^x$

vertical compression bafo $\frac{1}{2}$ $\left(\frac{y}{2}\right)$

b) $y = 8^{4x}$

- horizontal compression bafo $\frac{1}{4}$ $\left(\frac{x}{4}\right)$

c) $y = -8^x$

- vertical reflection, $(-y)$

d) $y = 8^{-2x}$

- horizontal compression bafo $\frac{1}{2}$ and horizontal reflection $\left(\frac{x}{-2}\right)$

5) Write the equation for the function that results from each transformation applied to the base function $y = 7^x$

a) reflect in the x-axis (vertical reflection)

$y = -7^x$

b) stretch vertically by a factor of 3

$y = 3(7)^x$

c) stretch horizontally by a factor of 2.4

$y = 7^{\frac{1}{2.4}x}$

d) reflect in the y-axis and ~~compress~~ stretch vertically by bafo 7

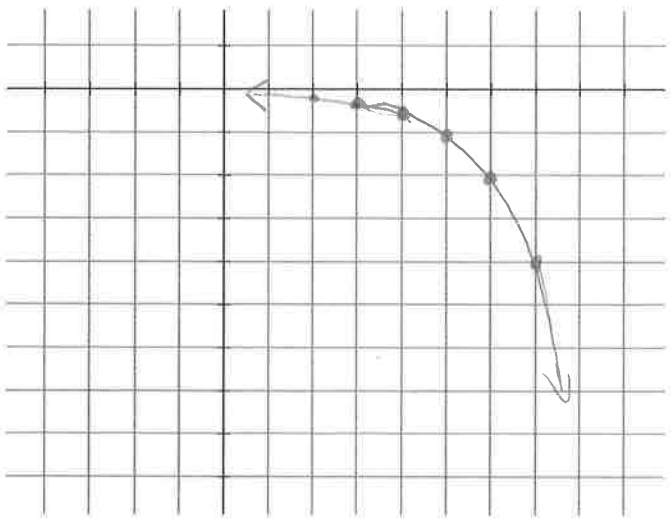
$y = 7(7)^{-x}$

6) Sketch the graph of $y = \left(-\frac{1}{2}\right)2^{x-4}$ by using $y = 2^x$ as the base and applying transformations.

$y = 2^x$

x	y
-3	$\frac{1}{8}$
-2	$\frac{1}{4}$
-1	$\frac{1}{2}$
0	1
1	2
2	4
3	8

$x+4$	$\frac{y}{-2}$
1	-0.0625
2	-0.125
3	-0.25
4	-0.5
5	-1
6	-2
7	-4



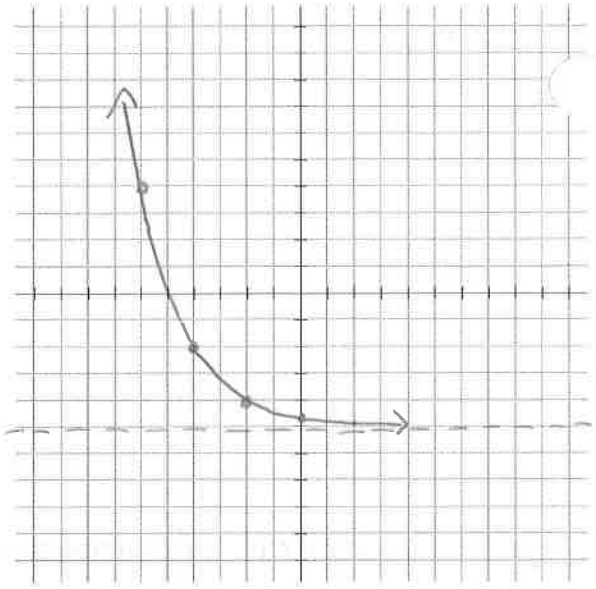
7) Sketch the graph of $y = 3^{-0.5x-1} - 5$ by using $y = 3^x$ as the base and applying transformations.

$y = 3^x$

x	y
-3	1/27
-2	1/9
-1	1/3
0	1
1	3
2	9
3	27

$y = 3^{-0.5(x+2)} - 5$

$-2x-2$	$y-5$
4	-4.96
2	-4.88
0	-4.67
-2	-4
-4	-2
-6	4
-8	22



8) a) Rewrite $y = 9^x$ using a base of 3. Describe how you can graph this function by transforming the graph of $y = 3^x$.

$y = 9^x$

$= (3^2)^x$

$= 3^{2x}$

horizontal compression bafo $\frac{1}{2}$

b) Rewrite $y = 9^x$ using a base of 81. Describe how you can graph this function by transforming the graph of $y = 81^x$.

$y = 9^x$

$= (81^{1/2})^x$

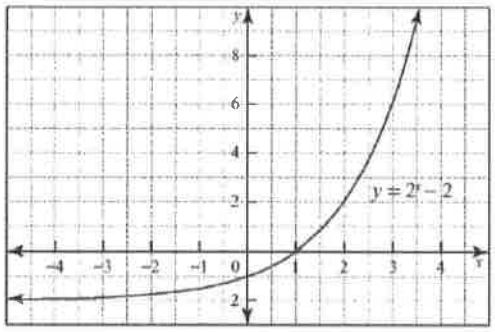
$= 81^{\frac{1}{2}x}$

horizontal stretch bafo 2.

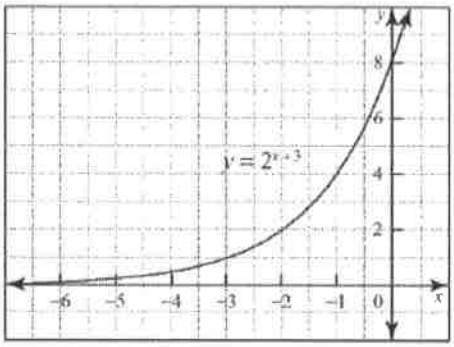
Answers

1) a) translate 2 units down **b)** translate 3 units left **c)** horizontal compression by a factor of $\frac{1}{2}$
d) vertical stretch by a factor of 3, a translation 1 unit to the right and 1 unit up

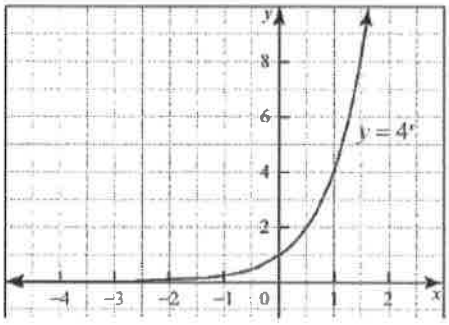
2) a)



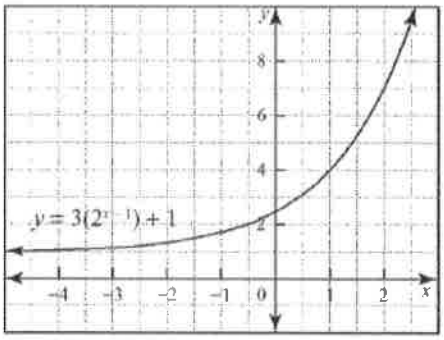
b)



c)



d)

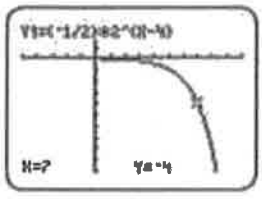


3) a) $y = 5^x - 3$ **b)** $y = 5^{x-2}$ **c)** $5^{x+\frac{1}{2}}$ **d)** $y = 5^{x+2.5} + 1$

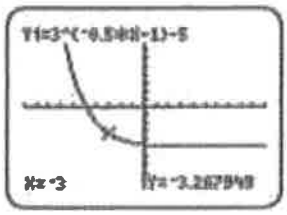
4) a) vertical compression by a factor of $\frac{1}{2}$ **b)** horizontal compression by a factor of $\frac{1}{4}$
c) vertical reflection (reflection in the x-axis)
d) horizontal reflection (reflection in the y-axis) and horizontal compression by a factor of $\frac{1}{2}$

5) a) $y = -7^x$ **b)** $y = 3(7^x)$ **c)** $y = 7^{\frac{x}{2.4}}$ **d)** $y = 7(7^{-x})$

6)



7)



8) a) $y = 3^{2x}$; horizontal compression of the graph of $y = 3^x$ by a factor of $\frac{1}{2}$

b) $y = 81^{\frac{1}{2}x}$; horizontal stretch of the graph of $y = 81^x$ by a factor of 2